Muhammad Mahasin Irfan

mahasin.irfan@gmail.com | Lahore, Pakistan | github.com/maani3 | linkedin.com/in/mahasin-irfan

Research Interests

Efficient and trustworthy systems; performance measurement; data privacy; human centered security and privacy.

Education

Lahore University of Management Sciences (LUMS) Lahore, Pakistan

B.S. in Computer Science; CGPA: 3.85/4.00 *September 2023 – May 2027 (expected)*

- Dean's Honor List in every completed academic year (2023–24, 2024–25, 2025–26).

Workshop Paper

M. M. Irfan, M. S. Hassan, and F. Zaffar. "Decomposing Error in Continual World Models." **NeurIPS 2026 Workshop on Continual World Models**, Idea Track. *Accepted*. PDF.

Selected Research Experience

Heterogeneous GPU/PIM Workload Splitting November 2025 – Present

Undergraduate Researcher, LUMS *Lahore, Pakistan*

Advisors/Collaborators: Dr. Fareed Zaffar; Humza Ikram (CMU); Dr. Hashim Sharif (AMD Research)

- Built repeatable GPU/PIM measurement pipelines across machines, including scripts and Docker environments; evaluated configurations on four CNNs and four LLMs.
- Extended the Newton PIM model in Ramulator using C++ to test additional bank groups and wider rows with corresponding t_{FAW} changes; measured 1.3× and 1.5× simulated throughput improvements over the original configuration.
- Developed a prediction guided split search that evaluates up to 7–8× fewer candidates while selecting configurations within 1% of exhaustive oracle end-to-end latency in the best evaluated cases. Tested limitations on nonmonotonic TensorRT curves.

Evaluation of Continual World Model Updates July 2026 – Present

First Author Researcher, LUMS *Lahore, Pakistan*

Advisors/Collaborators: Dr. Fareed Zaffar

- Derived an exact decomposition of post-update prediction error into inherited error and update mismatch, and wrote the first draft of the accepted workshop paper.
- Designed controlled experiments across prediction horizons; found opposite endpoint/mismatch rankings in 544/544 retained one-step linear trials and reproduced the phenomenon across 500 learned-model runs.

Security and Privacy in Mobile Phone Shops June 2026 – Present

Joint First Author Researcher, LUMS/CIRCLE *Lahore, Pakistan*

Advisors/Collaborators: Dr. Mobin Javed; Dr. Katharina Krombholz (CISPA); Dr. Lea Grober (ICSI Berkeley; formerly CISPA)

- Co-designed and led fieldwork for an IRB-approved qualitative study of how shopkeepers support account recovery, authentication, device transfers, and privacy for low-income communities in Pakistan.
- Conducted 15 semi-structured Urdu interviews in Lahore; transcribed, translated, and coded the interviews and contributed to collaborative reflexive thematic analysis. Manuscript in preparation.

Teaching Experience

Teaching Assistant, Computer Systems and Assembly Language (CS225) Fall 2026

LUMS; instructors: Dr. Zartash Afzal Uzmi and Dr. Jahangir *~350 students*

- Develop quizzes, assignments, and a new systems laboratory; grade examinations and conduct tutorials.

Teaching Assistant, Foundations of Data Science Spring 2026

LUMS; instructor: Dr. Mobin Javed *~200 students*

- Designed Python and data visualization assignments, automated tests, and Jupyter/Otter-Grader workflows.

Teaching Assistant, Introduction to Programming (CS100) Spring 2026

LUMS; instructor: Dr. Fareed Zaffar *~180 students*

- Created assignments, held programming labs and a tutorial, and provided office hours.

Academic Service

Selected Reviewer, NeurIPS 2026 Workshop on Continual World Models 2026

Selected by the workshop Technical Program Committee

- Reviewed one anonymous submission and sent a written review and recommendation to the program committee.

Technical and Research Skills

Python, C++, C, Linux, Docker, Git, PyTorch, Ramulator, TVM, performance benchmarking, simulation, semi-structured interviewing, reflexive thematic analysis; Urdu and Punjabi fieldwork.

Selected Honors

Dean's Honor List (all completed years) | National Mathematics Talent Contest, Pakistan top 20 | SEAMO Gold Medal, third in Pakistan | Purple Comet! Math Meet, first-place team in Pakistan